Topics

- Power series
- Alternating series test
- Conditionally vs. Absolutely convergent series
- Rearrangements

Def Given a sequence (c_n) of real numbers called "coefficients", we get a series

$$\sum_{n=0}^{\infty} c_n x^n$$

for every $x \in \mathbb{R}$ called a <u>power series</u>.

For some $x \in \mathbb{R}$ this will converge, for some $x \in \mathbb{R}$ it will not.

Thm Given $\sum c_n x^n$, let $\alpha = \limsup \sqrt[n]{c_n}$ and let $R = \frac{1}{\alpha}$.

(If $\alpha = 0$, $R = +\infty$, if $\alpha = +\infty$, $R = 0$.)

(a) $\sum c_n x^n$ converges for $|x| < R$

(b) $\sum c_n x^n$ diverges for $|x| > R$

Def R is the <u>radius of convergence</u> of $\sum c_n x^n$.

Proof Apply root test:

$$\limsup \sqrt[n]{|c_n x^n|} = \limsup |x| \sqrt[n]{|c_n|} = |x| \alpha.$$

$|x| < \frac{1}{\alpha} \Rightarrow$ convergence

$|x| > \frac{1}{\alpha} \Rightarrow$ divergence

Examples Find the radius of convergence of the following power series. What happens when $|x|$ equals radius of conv?

(a) $\sum n^n z^n$

(b) $\sum \frac{z^n}{n!}$

(c) $\sum z^n$

(d) $\sum \frac{z^n}{n}$

(e) $\sum \frac{z^n}{n^2}$

(a) $\sqrt[n]{n^n} = n \to \infty$ $\qquad R = 0$

(b) $\dfrac{z^{n+1}}{(n+1)!} \dfrac{n!}{z^n} = \dfrac{z}{n+1} \to 0$ $\qquad R = \infty$

(c) $\sqrt[n]{1} = 1 \to 1$ $\qquad R = 1$

for $x=1$ div.

for $x=-1$ div.

(d) $\sqrt[n]{\dfrac{1}{n}} \to 1$ $\qquad R = 1$

for $x=1$ div.

for $x=-1$ conv.

(e) $\sqrt[n]{\dfrac{1}{n^2}} \to 1$ $\qquad R = 1$

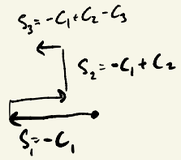
for $x=1$ conv.

for $x=-1$ conv.

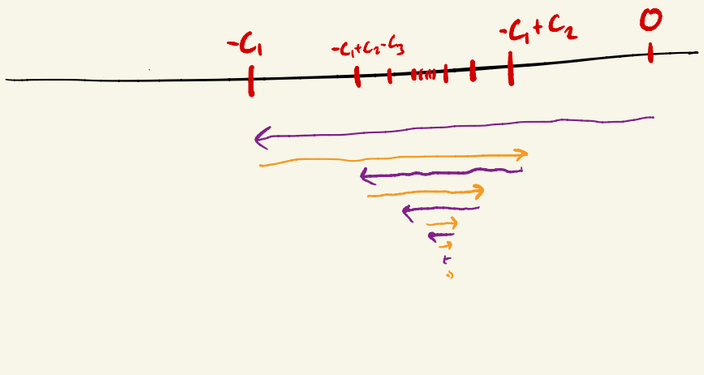
Thm (3.43) [Alternating series test] Suppose

(a) $c_1 \geq c_2 \geq \cdots$

(b) $\lim_{n\to\infty} c_n = 0$



Then $\sum_{n=0}^{\infty} (-1)^n c_n$ converges.



Proof: Let $S_m = c_1 + \cdots + c_m$. S_{2m} is increasing:

$$S_{2m+1} = S_{2m-1} + \underbrace{c_{2m} - c_{2m+1}}_{\text{positive}}$$

Similarly, S_{2m} is decreasing.

Also, $S_{2m-1} \leq S_{2m} \leq S_2$, so (S_{2m-1}) is bounded above, thus converges to some S.

For all ε, $\exists N$ s.t. $c_{2n} < \frac{\varepsilon}{2}$ and $|S_{2n-1} - S| < \frac{\varepsilon}{2}$ for $n \geq N$.

We have

$$|S_{2n} - S| = |S_{2n-1} + C - S| < \varepsilon$$

for all $n \geq N$ as well, so $\lim_{m \to \infty} S_m = S$.

Absolute vs. Conditional Convergence

Def $\sum a_n$ **converges absolutely** if $\sum |a_n|$ converges.

Thm If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.

proof: Cauchy criterion: $\left|\sum_{k=n}^{m} a_k\right| < \sum_{k=n}^{m} |a_k|$

Def If $\sum a_n$ converges, but not absolutely, it is **conditionally convergent.** (or "non-absolutely")

Example

$\sum \frac{(-1)^n}{n}$ is conditionally convergent.

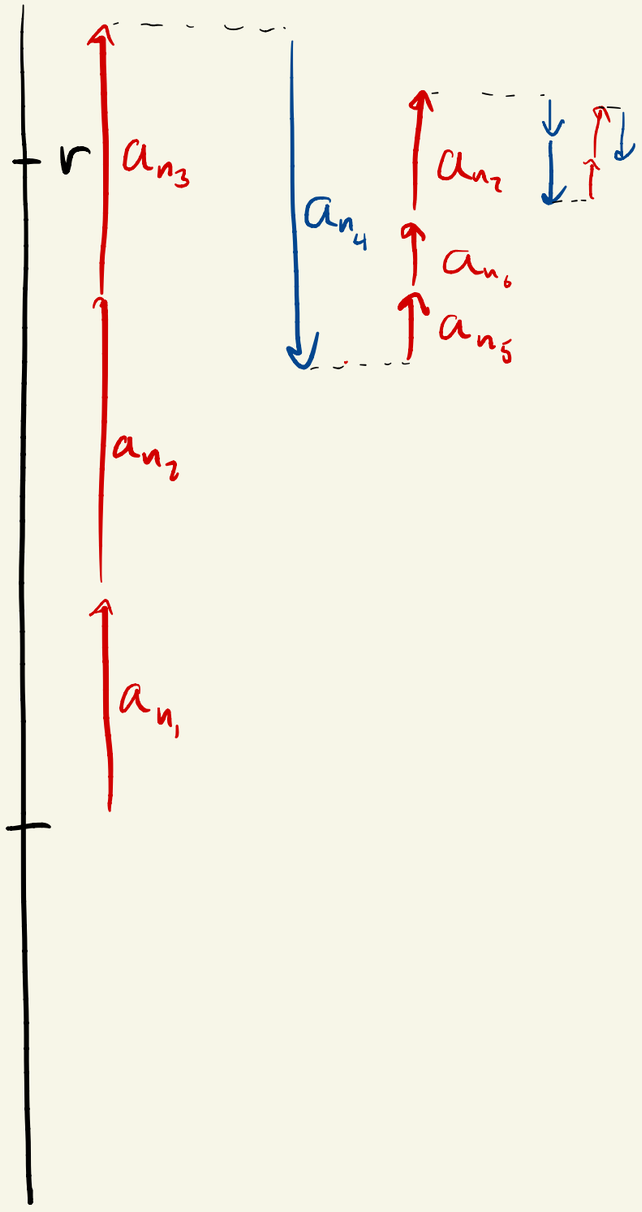
Absolutely convergent series behave well, like finite sums.

Conditionally convergent series, not so much.

Rearrangements

Def Let n_k be a sequence in $\mathbb{N}$ so that every natural number appears exactly once, i.e. $k \mapsto n_k$ is a bijection $\mathbb{N} \to \mathbb{N}$. $\sum_{k=1}^{\infty} a_{n_k}$ is a <u>rearrangement</u> of $\sum_{n=1}^{\infty} a_n$.

Thm Suppose $\sum a_n$ is convergent, but $\sum |a_n|$ is not. (i.e. $\sum a_n$ is <u>conditionally</u> convergent.) Then for any $r \in \mathbb{R}$ there is a rearrangement $\sum a_{n_k}$ s.t. $\sum_{k=1}^{\infty} a_{n_k} = r$.



Proof outline:

① There must be ∞ many positive and negative a_n and both must be divergent series.

② Choose $a_{n_1}, \dots, a_{n_{k_1}}$ to be the first k_1 positive terms where k_1 is the smallest # s.t. $a_{n_1} + \cdots + a_{n_{k_1}} > r$

③ Choose $a_{n_{k_1+1}}, \dots, a_{n_{k_2}}$ to be just enough negative terms s.t. $a_{n_1} + \cdots + a_{k_2} < r$.

④ Repeat steps ② and ③ forever using induction.

⑤ Show the rearrangement gives r.

Thm (3.55) If $\sum a_n$ is absolutly convergent, and $\sum a_{n_k}$ is a rearrangement, then $\sum a_{n_k}$ is convergent, and $\sum a_{n_k} = \sum a_n$.

Proof Given $\varepsilon > 0$, $\exists$ N s.t. $N \le n \le m$ implies

$$\sum_{k=n}^{m} |a_k| \le \varepsilon$$

Let N' be s.t. $\{1, 2, \dots, N\} \subset \{n_1, n_2, \dots, n_{N'}\}$.

If $n \ge N'$, then $\left| \sum_{k=1}^{n} a_k - \sum_{k=1}^{n} a_{n_k} \right|$ is a sum of finitely many $|a_k|$ with $k > N$, so is less than ε.